



COMPETENCY STANDARD

FOR

IC DESIGN AND LAYOUT ENGINEERING

(SEMICONDUCTOR / VLSI SECTOR)

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for IC Design and Layout Engineering serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the Semiconductor/ VLSI sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for IC Design and Layout Engineering. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level-15 & 16)
71-72 Old Elephant Road, Eskaton Garden
Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: <https://sicip.gov.bd/>

INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:
Generic Competencies (08 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-ICDE-01-G	Apply OHS, ESD and Safe Workplace Practices	1. Identify OHS and ESD requirements 2. Practice safe workstation and EDA lab procedures 3. Follow emergency response procedures	04
SICIP-SCD-ICDE-02-G	Apply Workplace Communication and Professional Ethics	1. Interpret workplace communication and etiquette 2. Prepare technical documents 3. Participate in design review and presentation	04
Total Hour			08 Hrs.

Sector Specific Competencies (32 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-ICDE-01-S	Interpret VLSI Industry and Semiconductor Fundamentals	1. Recognize VLSI industry and job opportunities. 2. Interpret chip design flow. 3. Identify industry readiness requirements. 4. Explain semiconductor devices. 5. Interpret CMOS fabrication steps. 6. Analyze device and process considerations.	16
SICIP-SCD-ICDE-02-S	Operate Linux and EDA Tools	1. Operate Linux command line. 2. Manage user environment and process tasks. 3. Organize EDA project files. 4. Set up EDA tool and library environment. 5. Capture schematics and run simulation. 6. Interpret simulation results.	16
Total Hour			32 Hrs.

Occupation Specific Competencies (200 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-ICDE-01-O	Design, Layout and Verify CMOS Standard Cells	1. Apply digital logic fundamentals. 2. Design and simulate CMOS inverter and logic cells. 3. Prepare standard-cell schematic deliverables. 4. Prepare layout plan and device structures. 5. Create standard-cell layouts. 6. Verify and debug layouts.	48
SICIP-SCD-ICDE-02-O	Design, Layout and Verify Sequential and Arithmetic Circuits	1. Design sequential circuits. 2. Design arithmetic circuits. 3. Complete layout and post-layout verification.	32
SICIP-SCD-ICDE-03-O	Analyze SRAM Memory Architecture and 6T Bit-Cell Operation	1. Interpret memory architecture. 2. Analyze 6T SRAM bit-cell operation. 3. Interpret SRAM peripherals and stability.	24
SICIP-SCD-ICDE-04-O	Design, Layout and Verify Analog Circuits	1. Apply analog design fundamentals. 2. Design current mirror and differential pair. 3. Design operational amplifiers. 4. Apply analog layout techniques. 5. Create matched analog layouts. 6. Perform post-layout verification.	52
SICIP-SCD-ICDE-05-O	Address Parasitic Effects, Reliability and Advanced Process Considerations	1. Identify parasitic and reliability effects. 2. Apply prevention and mitigation techniques. 3. Recognize advanced process structures.	16
SICIP-SCD-ICDE-06-O	Perform Analog IP Layout Project and Final Assessment	1. Plan and execute the project. 2. Prepare documentation and portfolio. 3. Present and defend design work.	28
Total Hour			200 Hrs.

COMPETENCY STANDARD: IC DESIGN AND LAYOUT ENGINEERING

The Generic Competencies

Unit of Competency: APPLY OHS, ESD AND SAFE WORKPLACE PRACTICES	Nominal Duration: 04 hrs.	Unit Code: SICIP-SCD-ICDE-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply OHS, ESD and Safe Workplace Practices. It specifically includes the tasks of identifying OHS and ESD requirements, practicing safe workstation and EDA lab procedures and following emergency response procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS and ESD requirements	1.1 <u>Workplace OHS and ESD requirements</u> are identified according to organizational procedures. 1.2 <u>Hazards</u> associated with semiconductor and EDA laboratory environments are recognized. 1.3 <u>Personal protective equipment (PPE) and ESD protection devices</u> are identified and used appropriately. 1.4 ESD control measures are applied to protect electronic components and equipment. 1.5 <u>Workplace safety signs, symbols, and instructions</u> are interpreted correctly.
2. Practice safe workstation and EDA lab procedures	2.1 <u>Workstations</u> are prepared according to OHS and ESD requirements. 2.2 <u>EDA tools, computers, and laboratory equipment</u> are operated following safe work procedures. 2.3 <u>Cables, electrical equipment, and work areas</u> are maintained to prevent workplace hazards. 2.4 Good housekeeping practices are maintained to ensure a safe working environment. 2.5 Workplace incidents, hazards, and unsafe conditions are identified and reported promptly.
3. Follow emergency response procedures	3.1 <u>Emergency procedures</u> are identified according to workplace requirements. 3.2 Emergency exits, assembly points, and safety equipment are located and accessed correctly. 3.3 Appropriate actions are taken during electrical, fire, or other workplace emergencies. 3.4 Incidents and accidents are reported according to organizational procedures.

	3.5 <u>Emergency response activities</u> are carried out in accordance with workplace safety requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Workplace OHS and ESD requirements	1.1 OHS Policies 1.2 ESD Procedures 1.3 Workplace Rules 1.4 Safe Work Practices 1.5 Emergency Procedures 1.6 Risk Assessment
2. Hazards	2.1 Electrostatic Discharge (ESD) 2.2 Electrical Hazards 2.3 Fire Hazards 2.4 Chemical Hazards 2.5 Trip and Slip Hazards 2.6 Equipment Hazards
3. Personal protective equipment (PPE) and ESD protection devices	3.1 ESD Wrist Strap 3.2 ESD Shoes 3.3 ESD Mat 3.4 Safety Glasses 3.5 Protective Gloves 3.6 Lab Coat
4. Workplace safety signs, symbols, and instructions	4.1 Warning Signs 4.2 Mandatory Signs 4.3 Prohibition Signs 4.4 Emergency Exit Signs 4.5 Fire Safety Signs 4.6 Hazard Symbols
5. Workstations	5.1 ESD Workstation 5.2 Computer Workstation 5.3 Design Workbench 5.4 Laboratory Bench 5.5 Test Workstation 5.6 Simulation Workstation
6. EDA tools, computers, and laboratory equipment	6.1 EDA Software 6.2 Workstation Computer 6.3 Server 6.4 Oscilloscope 6.5 Digital Multimeter 6.6 Power Supply
7. Cables, electrical equipment, and work areas	7.1 Power Cables 7.2 Network Cables 7.3 Extension Cords 7.4 Electrical Outlets

	7.5 Work Benches 7.6 Cable Management
8. Emergency procedures	8.1 Fire Emergency 8.2 Electrical Emergency 8.3 Medical Emergency 8.4 Evacuation Procedure 8.5 First Aid Procedure 8.6 Incident Reporting Procedure
9. Emergency response activities	9.1 Evacuation 9.2 First Aid 9.3 Firefighting 9.4 Hazard Isolation 9.5 Emergency Notification 9.6 Incident Documentation

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Principles of Occupational Health and Safety (OHS) and Electrostatic Discharge (ESD) protection. 1.2 Common hazards in semiconductor fabrication, EDA laboratories, and electronic work environments. 1.3 Proper use of Personal Protective Equipment (PPE) and ESD protection devices. 1.4 ESD control methods and safe handling of electronic components. 1.5 Safe operation of computers, EDA tools, electrical equipment, and laboratory workstations. 1.6 Workplace housekeeping practices and electrical safety requirements. 1.7 Emergency procedures, including fire safety, electrical hazards, evacuation routes. 1.8 Interpretation of workplace safety signs, symbols, and organizational safety policies and procedures.
2. Underpinning Skills	2.1 Applying workplace OHS and ESD requirements in semiconductor and EDA laboratory environments 2.2 Interpreting workplace safety signs, symbols, and safety instructions 2.3 Preparing and maintaining safe workstations and EDA laboratory environments 2.4 Operating EDA tools, computers, and laboratory equipment safely 2.5 Applying good housekeeping practices to maintain a hazard-free workplace 2.6 Following emergency response procedures for electrical, fire, and workplace emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities

	3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning/operational manual 4.3 PPEs 4.4 Firefighting equipment 4.5 Emergency response manual 4.6 Emergency Systems (Fire alarms, emergency exits, etc.) 4.7 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified OHS and ESD requirements 1.2 practiced safe workstation and EDA lab procedures 1.3 followed emergency response procedures
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY WORKPLACE COMMUNICATION AND PROFESSIONAL ETHICS	Nominal Duration: 04 hrs.	Unit Code: SICIP-SCD-ICDE-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply workplace communication and professional ethics. It specifically includes the tasks of interpreting workplace communication and etiquette, preparing technical documents and participating in design review and presentation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 <u>Workplace communication methods</u> and organizational protocols are identified. 1.2 <u>Professional etiquette</u> is demonstrated during workplace interactions. 1.3 Technical information and work instructions are interpreted accurately. 1.4 <u>Verbal and written communication</u> is conducted clearly and professionally. 1.5 Team communication and collaboration are maintained according to workplace requirements.
2. Prepare technical documents	2.1 <u>Technical information</u> is collected and organized according to project requirements. 2.2 Design notes, reports, and technical documents are prepared using standard formats. 2.3 <u>Technical terminology and documentation standards</u> are applied correctly. 2.4 Documents are reviewed for accuracy, completeness, and consistency. 2.5 Final documents are submitted according to organizational procedures.
3. Participate in design review and presentation	3.1 <u>Design data and supporting materials</u> are prepared for review meetings. 3.2 Design concepts and technical information are presented clearly to stakeholders. 3.3 Feedback and review comments are recorded and analyzed. 3.4 <u>Design modifications</u> are incorporated based on review outcomes. 3.5 Professional communication and teamwork are demonstrated throughout the review and presentation process.

Range of Variables

Variable	Range May Include but not limited to:
1. Workplace communication methods	1.1 Face-to-Face Communication 1.2 Email 1.3 Phone Calls 1.4 Video Conferencing 1.5 Instant Messaging
2. Professional etiquette	2.1 Respectful Behaviour 2.2 Professional Language 2.3 Punctuality 2.4 Active Listening 2.5 Appropriate Dress Code 2.6 Workplace Ethics
3. Verbal and written communication	3.1 Meetings 3.2 Emails 3.3 Reports 3.4 Presentations 3.5 Memos 3.6 Technical Documentation
4. Technical information	4.1 Design Specifications 4.2 Project Requirements 4.3 Technical Drawings 4.4 Test Results 4.5 Design Data 4.6 Reference Documents
5. Technical terminology and documentation standards	5.1 Technical Terms 5.2 Industry Standards 5.3 Naming Conventions 5.4 Document Templates 5.5 File Formats 5.6 Documentation Guidelines
6. Design data and supporting materials	6.1 Design Drawings 6.2 Schematics 6.3 Presentation Slides 6.4 Technical Reports 6.5 Simulation Results 6.6 Design Specifications
7. Design modifications	7.1 Design Updates 7.2 Specification Changes 7.3 Layout Revisions 7.4 Documentation Updates 7.5 Error Corrections 7.6 Improvement Actions

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Principles of workplace communication, professional etiquette, and organizational communication protocols. 1.2 Methods for interpreting technical information, work instructions, and engineering documentation. 1.3 Standards and formats for preparing technical reports, design notes, and project documentation. 1.4 Use of technical terminology, documentation conventions, and version control practices. 1.5 Principles of effective presentation, technical discussion, and design review processes. 1.6 Methods for receiving, incorporating technical feedback into design improvements. 1.7 Teamwork, collaboration, and professional communication practices in engineering.
2. Underpinning Skills	2.1 Applying professional workplace communication and organizational etiquette 2.2 Collaborating with team members to support design and project activities 2.3 Preparing technical documents, reports, and design notes using standard formats 2.4 Applying technical terminology and documentation standards correctly 2.5 Reviewing technical documents for accuracy, completeness, and consistency 2.6 Presenting design concepts and technical information clearly during review meetings 2.7 Incorporating feedback to improve design quality and project outcomes
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Work books 4.3 Learning manuals 4.4 Technical documents

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted workplace communication and etiquette 1.2 prepared technical documents 1.3 participated in design review and presentation
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Sector Specific Competencies

Unit of Competency: INTERPRET VLSI INDUSTRY AND SEMICONDUCTOR FUNDAMENTALS	Nominal Duration: 16 hrs.	Unit Code: SICIP-SCD-ICDE-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret VLSI industry and Semiconductor Fundamentals. It specifically includes the tasks of recognizing VLSI industry and job opportunities, interpreting chip design flow, identifying industry readiness requirements, explaining semiconductor devices, interpreting CMOS fabrication steps and analyzing device and process considerations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Recognize VLSI industry and job opportunities.	1.1 <u>VLSI industry structure and application areas</u> are identified. 1.2 <u>Semiconductor sectors</u> are distinguished. 1.3 Common job roles and career pathways in the VLSI industry are identified. 1.4 Skills and competencies required for VLSI careers are recognized. 1.5 Industry trends and employment opportunities are interpreted.
2. Interpret chip design flow.	2.1 <u>Complete chip design flow</u> from specification to fabrication is interpreted. 2.2 Front-end and back-end design activities are distinguished. 2.3 Major design stages, tools, and deliverables are identified. 2.4 <u>Design verification</u> and physical implementation processes are explained. 2.5 Design flow documentation is interpreted according to project requirements.
3. Identify industry readiness requirements.	3.1 <u>Technical knowledge and workplace competencies</u> required for the VLSI industry are identified. 3.2 Industry standards, documentation, and quality requirements are recognized. 3.3 Workplace communication, teamwork, and professional ethics are identified. 3.4 <u>Continuous learning and skill development opportunities</u> are recognized.

	3.5 Industry expectations and workplace practices are interpreted.
4. Explain semiconductor devices.	4.1 <u>Semiconductor materials</u> and their characteristics are explained. 4.2 PN junction, diode, and transistor operating principles are described. 4.3 <u>MOSFET structure and operation</u> are explained. 4.4 Basic electronic characteristics of semiconductor devices are interpreted. 4.5 <u>Semiconductor devices</u> are related to digital IC applications.
5. Interpret CMOS fabrication steps.	5.1 <u>CMOS fabrication flow</u> is interpreted from wafer preparation to packaging. 5.2 Major fabrication processes and metallization are identified. 5.3 Fabrication terminology and process sequence are interpreted correctly. 5.4 Cleanroom practices and process control requirements are recognized. 5.5 <u>Fabrication outputs</u> are interpreted according to process requirements.
6. Analyze device and process considerations.	6.1 <u>Device characteristics affecting IC performance</u> are analyzed. 6.2 Process variations and manufacturing limitations are identified. 6.3 Power, performance, and area (PPA) considerations are interpreted. 6.4 <u>Reliability and yield factors</u> are analyzed according to design requirements. 6.5 Device and process considerations are evaluated for successful IC implementation.

Range of Variables

Variable	Range
	May include but not limited to:
1. VLSI industry structure and application areas	1.1 IC Design 1.2 Wafer Fabrication 1.3 Packaging and Assembly 1.4 Semiconductor Testing 1.5 Consumer Electronics 1.6 Automotive and Industrial Electronics
2. Semiconductor sectors	2.1 IC Design 2.2 Wafer Fabrication 2.3 Foundry 2.4 Packaging 2.5 Assembly

	2.6 Testing
3. Complete chip design flow	3.1 Design Specification 3.2 RTL Design 3.3 Functional Verification 3.4 Logic Synthesis 3.5 Physical Design 3.6 Tape-out and Fabrication
4. Design verification	4.1 Functional Simulation 4.2 Static Timing Analysis (STA) 4.3 Design Rule Check (DRC) 4.4 Layout Versus Schematic (LVS)
5. Technical knowledge and workplace competencies	5.1 Digital Electronics 5.2 Verilog/SystemVerilog 5.3 Linux and Shell Scripting 5.4 EDA Tools 5.5 Static Timing Analysis (STA) 5.6 Problem-Solving Skills
6. Continuous learning and skill development opportunities	6.1 Technical Training 6.2 Professional Certifications 6.3 Workshops and Seminars 6.4 Online Learning Platforms 6.5 Industry Conferences 6.6 Vendor Training Programs
7. Semiconductor materials	7.1 Silicon (Si) 7.2 Germanium (Ge) 7.3 Gallium Arsenide (GaAs) 7.4 Intrinsic Semiconductor 7.5 N-type Semiconductor 7.6 P-type Semiconductor
8. MOSFET structure and operation	8.1 NMOS 8.2 PMOS 8.3 Gate 8.4 Source 8.5 Drain 8.6 Channel
9. Semiconductor devices	9.1 CMOS Logic Gates 9.2 Memory Circuits 9.3 Microprocessors 9.4 ASICs 9.5 FPGAs 9.6 System-on-Chip (SoC)
10. CMOS fabrication flow	10.1 Wafer Preparation 10.2 Oxidation 10.3 Photolithography 10.4 Etching 10.5 Doping (Ion Implantation) 10.6 Packaging

11. Fabrication outputs	11.1 Processed Wafer 11.2 Wafer Map 11.3 Yield Report 11.4 Inspection Report 11.5 Packaged IC 11.6 Process Documentation
12. Device characteristics affecting IC performance	12.1 Threshold Voltage (V_{th}) 12.2 Switching Speed 12.3 Leakage Current 12.4 Drive Strength 12.5 Propagation Delay 12.6 Power Consumption
13. Reliability and yield factors	13.1 Electromigration (EM) 13.2 Bias Temperature Instability (BTI) 13.3 IR Drop 13.4 Yield 13.5 Defect Density 13.6 Process Reliability

Curricular Content Guide

1. Underpinning Knowledge	1.1 VLSI industry structure, semiconductor sectors, and career opportunities. 1.2 Complete chip design flow from specification to fabrication. 1.3 Front-end and back-end VLSI design concepts and EDA tools. 1.4 Semiconductor device fundamentals, including diode, transistor, and MOSFET operation. 1.5 CMOS fabrication process, cleanroom practices, and manufacturing basics. 1.6 Industry standards, workplace practices, communication, and professional ethics. 1.7 Power, Performance, and Area (PPA), process variation, reliability, and yield concepts. 1.8 Continuous learning, emerging VLSI technologies, and industry readiness skills.
2. Underpinning Skills	2.1 Recognizing industry readiness requirements, workplace competencies, and professional practices 2.2 Applying knowledge of semiconductor devices, including diodes, transistors, and MOSFETs 2.3 Interpreting CMOS fabrication processes from wafer preparation to packaging 2.4 Understanding cleanroom practices, fabrication terminology, and process control requirements 2.5 Analyzing semiconductor device characteristics and process variations affecting IC performance

	2.6 Evaluating power, performance, area (PPA), reliability, and yield considerations in IC design 2.7 Applying semiconductor and VLSI fundamentals to support successful chip design and implementation
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 recognized VLSI industry and job opportunities. 1.2 interpreted chip design flow. 1.3 identified industry readiness requirements. 1.4 explained semiconductor devices. 1.5 interpreted CMOS fabrication steps. 1.6 analyzed device and process considerations.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE LINUX AND EDA TOOLS	Nominal Duration: 16 hrs.	Unit Code: SICIP-SCD-ICDE-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate LINUX and EDA tools. It specifically includes the tasks of operating LINUX command line, managing user environment and process tasks, organizing EDA project files, setting up EDA tool and library environment, capturing schematics and run simulation and interpreting simulation results.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Operate Linux command line.	1.1 <u>Linux terminal</u> is accessed according to workplace requirements. 1.2 Linux commands are used to navigate directories and manage files. 1.3 <u>Files and directories</u> are created, copied, moved, and deleted using command-line utilities. 1.4 File permissions and access rights are managed according to project requirements. 1.5 <u>Command execution results</u> are verified before proceeding with design tasks.
2. Manage user environment and process tasks.	2.1 <u>User environment variables</u> are configured according to EDA tool requirements. 2.2 Running processes are monitored and managed using Linux commands. 2.3 Shell commands and utilities are used to support daily design activities. 2.4 Process logs and system messages are reviewed to identify issues. 2.5 <u>User environment settings</u> are verified before EDA tool execution.
3. Organize EDA project files.	3.1 <u>Project directories</u> are created according to organizational standards. 3.2 <u>Design files, libraries, and simulation data</u> are organized systematically. 3.3 File naming conventions and version control practices are followed. 3.4 Project data is backed up and maintained according to workplace procedures. 3.5 Project files are verified before design and simulation activities.
4. Set up EDA tool and library environment.	4.1 <u>EDA tools</u> are configured according to project requirements.

	4.2 Technology libraries and design kits are loaded into the design environment. 4.3 <u>Environment variables and library paths</u> are verified before tool execution. 4.4 Tool configuration is tested to ensure proper operation. 4.5 EDA environment is prepared for schematic design and simulation.
5. Capture schematics and run simulation.	5.1 <u>Circuit schematics</u> are created using the appropriate EDA tool. 5.2 <u>Components and device parameters</u> are selected according to design requirements. 5.3 Simulation settings are configured before execution. 5.4 Circuit simulation is performed using the appropriate EDA tool. 5.5 Simulation outputs are generated and saved for analysis.
6. Interpret simulation results.	6.1 <u>Simulation waveforms and output data</u> are reviewed. 6.2 Circuit behavior is analyzed against design specifications. 6.3 <u>Functional errors and performance issues</u> are identified from simulation results. 6.4 Design modifications are made based on simulation analysis. 6.5 Simulation results are verified and documented according to project requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1. Linux terminal	1.1 Bash Shell 1.2 Terminal Emulator 1.3 SSH Session 1.4 Local Terminal 1.5 Remote Terminal 1.6 Command Prompt
2. Files and directories	2.1 mkdir 2.2 cp 2.3 mv 2.4 rm 2.5 touch 2.6 rmdir
3. Command execution results	3.1 Standard Output 3.2 Error Messages 3.3 Exit Status 3.4 Command History

	3.5 Log Files 3.6 Verification Results
4. User environment variables	4.1 PATH 4.2 LD_LIBRARY_PATH 4.3 HOME 4.4 SHELL 4.5 Tool Environment Variables 4.6 License Environment Variables
5. User environment settings	5.1 Environment Variables 5.2 Shell Configuration Files 5.3 Tool Paths 5.4 License Settings 5.5 Working Directory 5.6 User Profile Configuration
6. Project directories	6.1 Project Root Directory 6.2 RTL Directory 6.3 Simulation Directory 6.4 Synthesis Directory 6.5 Physical Design Directory 6.6 Verification Directory
7. Design files, libraries, and simulation data	7.1 RTL Files 7.2 Library Files 7.3 Constraint Files 7.4 Netlist Files 7.5 Simulation Waveforms 7.6 Testbench Files
8. EDA tools	8.1 Cadence Virtuoso 8.2 Cadence Innovus 8.3 Cadence Genus 8.4 Synopsys Design Compiler 8.5 Synopsys PrimeTime 8.6 Siemens Questa
9. Environment variables and library paths	9.1 PATH 9.2 LD_LIBRARY_PATH 9.3 Tool Environment Variables 9.4 Library Paths 9.5 License Environment Variables 9.6 PDK Paths
10. Circuit schematics	10.1 Schematic Editor 10.2 Analog Circuit Schematic 10.3 Digital Circuit Schematic 10.4 Hierarchical Schematic 10.5 Symbol View 10.6 Cell View
11. Components and device parameters	11.1 MOSFET 11.2 Resistor 11.3 Capacitor

	11.4 Inductor 11.5 Voltage Source 11.6 Device Parameters
12.Simulation waveforms and output data	12.1 Voltage Waveforms 12.2 Current Waveforms 12.3 Timing Waveforms 12.4 Simulation Reports 12.5 Output Data Files 12.6 Measurement Results
13.Functional errors and performance issues	13.1 Logic Errors 13.2 Timing Violations 13.3 Power Issues 13.4 Signal Integrity Issues 13.5 Convergence Errors 13.6 Output Mismatches

Curricular Content Guide

1. Underpinning knowledge	1.1 Linux command-line operations, file management, and user environment configuration. 1.2 Linux process management, shell utilities, and basic system administration. 1.3 EDA project organization, file structure, naming conventions, and version control. 1.4 EDA tool setup, technology libraries, design kits, and environment configuration. 1.5 Schematic capture principles and circuit simulation using EDA tools. 1.6 Simulation setup, waveform generation, and output analysis techniques. 1.7 Methods for identifying functional errors and validating circuit performance. 1.8 Documentation of simulation results and project data management.
2. Underpinning skills	2.1 Operating Linux command-line tools for file, directory, and permission management 2.2 Managing Linux user environments, processes, and system resources for EDA applications 2.3 Organizing EDA project files, libraries, and simulation data using standard practices 2.4 Configuring EDA tools, technology libraries, and design environments for project execution 2.5 Running circuit simulations and generating output data for verification 2.6 Interpreting simulation waveforms and analyzing circuit behavior against design specifications 2.7 Documenting simulation results and maintaining organized project data for design activities

3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 LINUX 4.4 EDA tools

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 operated Linux command line. 1.2 managed user environment and process tasks. 1.3 organized EDA project files. 1.4 set up EDA tool and library environment. 1.5 captured schematics and run simulation. 1.6 interpreted simulation results.
2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-specific Competencies

Unit of Competency: DESIGN, LAYOUT AND VERIFY CMOS STANDARD CELLS	Nominal Duration: 48 hrs.	Unit Code: SICIP-SCD-ICDE-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to design, layout and verify CMOS standard cells. It specifically includes the tasks of applying digital logic fundamentals, designing and simulate CMOS inverter and logic cells, preparing standard-cell schematic deliverables, preparing layout plan and device structures, creating standard-cell layouts and verifying and debug layouts.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply digital logic fundamentals.	1.1 <u>Digital logic requirements</u> are interpreted from the design specification. 1.2 Boolean logic, truth tables, and logic gates are applied to circuit design. 1.3 <u>Combinational logic circuits</u> are developed according to functional requirements. 1.4 Logic functions are verified before circuit implementation. 1.5 Digital logic design is documented according to project requirements.
2. Design and simulate CMOS inverter and logic cells.	2.1 <u>CMOS inverter and basic logic cell schematics</u> are created using the appropriate EDA tool. 2.2 PMOS and NMOS devices are configured according to design requirements. 2.3 <u>Circuit simulation</u> is performed to verify functional behavior. 2.4 Simulation waveforms and performance results are analyzed. 2.5 Circuit design is refined based on simulation results.
3. Prepare standard-cell schematic deliverables.	3.1 <u>Standard-cell schematics</u> are prepared according to design specifications. 3.2 Device parameters, connectivity, and pin definitions are verified. 3.3 Schematic symbols and design views are generated using the EDA tool. 3.4 <u>Design files</u> are reviewed for completeness and correctness. 3.5 Schematic deliverables are prepared for layout implementation.

4. Prepare layout plan and device structures.	4.1 <u>Layout requirements and design rules</u> are reviewed before layout creation. 4.2 Device placement and routing strategy are planned according to standard-cell architecture. 4.3 PMOS, NMOS, wells, and substrate structures are arranged according to layout requirements. 4.4 Power rails, cell boundaries, and routing resources are planned. 4.5 <u>Layout plan</u> is verified before layout implementation.
5. Create standard-cell layouts.	5.1 <u>Standard-cell layouts</u> are created using the appropriate layout editor. 5.2 Device placement, routing, contacts, and interconnections are completed. 5.3 <u>Power and ground connections</u> are implemented according to layout requirements. 5.4 Layout design is optimized for area and manufacturability. 5.5 Completed layouts are prepared for physical verification.
6. Verify and debug layouts.	6.1 <u>Design Rule Check (DRC)</u> is performed to verify layout compliance. 6.2 Layout Versus Schematic (LVS) verification is executed to confirm design correctness. 6.3 DRC and LVS violations are identified and analyzed. 6.4 Layout errors are corrected and reverified. 6.5 Verified layouts are approved for standard-cell library integration.

Range of Variables

Variable	Range May include but not limited to:
1. Digital logic requirements	1.1 Design Specification 1.2 Functional Requirements 1.3 Input/Output Requirements 1.4 Logic Functions 1.5 Timing Requirements 1.6 Design Constraints
2. Combinational logic circuits	2.1 Multiplexer (MUX) 2.2 Demultiplexer (DEMUX) 2.3 Encoder 2.4 Decoder 2.5 Comparator 2.6 Arithmetic Logic Unit (ALU)
3. CMOS inverter and basic logic cell schematics	3.1 CMOS Inverter 3.2 NAND Gate 3.3 NOR Gate 3.4 XOR Gate

	3.5 XNOR Gate 3.6 Buffer
4. Circuit simulation	4.1 DC Analysis 4.2 Transient Analysis 4.3 AC Analysis 4.4 Corner Analysis 4.5 Monte Carlo Analysis 4.6 Functional Simulation
5. Standard-cell schematics	5.1 Inverter Cell 5.2 NAND Cell 5.3 NOR Cell 5.4 XOR Cell 5.5 Flip-Flop Cell 5.6 Buffer Cell
6. Design files	6.1 Schematic Files 6.2 Symbol Files 6.3 Library Files 6.4 Configuration Files 6.5 Design Reports 6.6 Verification Reports
7. Layout requirements and design rules	7.1 Design Rule Manual (DRM) 7.2 Process Design Kit (PDK) 7.3 Minimum Spacing Rules 7.4 Minimum Width Rules 7.5 Enclosure Rules 7.6 Layout Constraints
8. Layout plan	8.1 Floorplan 8.2 Placement Plan 8.3 Routing Plan 8.4 DRC Checklist 8.5 Layout Review 8.6 Implementation Plan
9. Standard-cell layouts	9.1 CMOS Inverter Layout 9.2 NAND Gate Layout 9.3 NOR Gate Layout 9.4 XOR Gate Layout 9.5 Flip-Flop Layout 9.6 Standard Cell Library
10. Power and ground connections	10.1 VDD Rail 10.2 VSS Rail 10.3 Power Straps 10.4 Power Contacts 10.5 Ground Network 10.6 Well Taps
11. Design Rule Check (DRC)	11.1 Minimum Width Check 11.2 Minimum Spacing Check 11.3 Enclosure Check

	11.4 Overlap Check 11.5 Antenna Check 11.6 DRC Report
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Curricular Content Guide

1. Underpinning knowledge	1.1 Digital logic fundamentals, Boolean algebra, logic gates, and combinational circuit design. 1.2 CMOS inverter, PMOS/NMOS operation, and standard logic cell design principles. 1.3 Schematic capture, circuit simulation, and waveform analysis using EDA tools. 1.4 Standard-cell schematic preparation, symbols, pin definitions, and design documentation. 1.5 Standard-cell layout planning, device placement, routing, and power rail design. 1.6 Layout design rules, area optimization, and manufacturability considerations. 1.7 Physical verification using DRC and LVS, and layout debugging techniques. 1.8 Standard-cell library integration and layout quality verification.
2. Underpinning skills	2.1 Applying digital logic fundamentals using Boolean algebra, logic gates, and combinational circuits 2.2 Designing and simulating CMOS inverter and basic logic cells using EDA tools 2.3 Analyzing simulation waveforms and refining circuit designs based on results 2.4 Planning standard-cell layouts according to design rules and layout architecture 2.5 Creating optimized layouts with proper device placement, routing, and power connections 2.6 Performing DRC and LVS verification to ensure layout accuracy and compliance 2.7 Debugging layout violations and preparing verified standard-cell designs for library integration
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Tools & equipment 4.4 Job specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 applied digital logic fundamentals. 1.2 designed and simulated CMOS inverter and logic cells. 1.3 prepared standard-cell schematic deliverables. 1.4 prepared layout plan and device structures. 1.5 created standard-cell layouts. 1.6 verified and debug layouts.
2 Methods of assessment	Methods of assessment should include: 2.1 Written Test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3 Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DESIGN, LAYOUT AND VERIFY SEQUENTIAL AND ARITHMETIC CIRCUITS	Nominal Duration: 32 hrs.	Unit Code: SICIP-SCD-ICDE-02-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to design, layout and verify sequential and arithmetic circuits. It specifically includes the tasks of designing sequential circuits, designing arithmetic circuits and completing layout and post-layout verification.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Design sequential circuits.	1.1 <u>Sequential circuit requirements</u> are interpreted from the design specification. 1.2 Flip-flops, latches, registers, and counters are designed using the appropriate EDA tool. 1.3 Circuit schematics are created and verified for connectivity and functionality. 1.4 <u>Functional simulation</u> is performed to validate sequential circuit behavior. 1.5 Sequential circuit design is reviewed before layout implementation.
2. Design arithmetic circuits.	2.1 <u>Arithmetic circuit requirements</u> are identified according to design specifications. 2.2 Adders, subtractors, multiplexers, and other <u>arithmetic circuits</u> are designed using the appropriate EDA tool. 2.3 Circuit schematics are simulated to verify functional operation. 2.4 Simulation results are analyzed and design issues are corrected. 2.5 Verified arithmetic circuit schematics are prepared for layout implementation.
3. Complete layout and post-layout verification.	3.1 <u>Layouts for sequential and arithmetic circuits</u> are created according to design rules. 3.2 Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification are performed. 3.3 Post-layout extraction and simulation are conducted to evaluate circuit performance. 3.4 <u>Layout and simulation errors</u> are identified and corrected. 3.5 Verified layouts and post-layout results are approved for final submission.

Range of Variables

Variable	Range May include but not limited to:
1. Sequential circuit requirements	1.1 Design Specification 1.2 Functional Requirements 1.3 Timing Requirements 1.4 Clock Requirements 1.5 Reset Requirements 1.6 Design Constraints
2. Functional simulation	2.1 Transient Analysis 2.2 Timing Simulation 2.3 Clock Waveforms 2.4 Reset Verification 2.5 Functional Verification 2.6 Simulation Report
3. Arithmetic circuit requirements	3.1 Design Specification 3.2 Functional Requirements 3.3 Performance Requirements 3.4 Timing Requirements 3.5 Area Constraints 3.6 Power Constraints
4. Arithmetic circuits	4.1 Half Adder 4.2 Full Adder 4.3 Subtractor 4.4 Multiplexer (MUX) 4.5 Decoder 4.6 Arithmetic Logic Unit (ALU)
5. Layouts for sequential and arithmetic circuits	5.1 Flip-Flop Layout 5.2 Register Layout 5.3 Counter Layout 5.4 Adder Layout 5.5 Multiplexer Layout 5.6 Arithmetic Logic Unit (ALU) Layout
6. Layout and simulation errors	6.1 DRC Violations 6.2 LVS Mismatches 6.3 Connectivity Errors 6.4 Timing Violations 6.5 Parasitic Effects 6.6 Layout Corrections

Curricular Content Guide

1. Underpinning knowledge	1.1 Fundamentals of sequential circuits, including flip-flops, latches, registers, and counters. 1.2 Principles of arithmetic circuit design, including adders,
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	subtractors, and multiplexers. 1.3 Schematic capture, circuit simulation, and functional verification using EDA tools. 1.4 Layout design principles for sequential and arithmetic circuits. 1.5 Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification concepts. 1.6 Post-layout extraction, parasitic effects, and circuit performance analysis. 1.7 Techniques for debugging design, layout, and simulation errors. 1.8 Preparation and verification of final layout deliverables for submission.
2. Underpinning skills	2.1 Designing sequential circuits using flip-flops, latches, registers, and counters 2.2 Designing arithmetic circuits including adders, subtractors, multiplexers, and related logic 2.3 Creating and verifying circuit schematics using appropriate EDA tools 2.4 Performing functional simulation to validate sequential and arithmetic circuit behavior 2.5 Performing DRC and LVS verification to ensure layout correctness and compliance 2.6 Conducting post-layout extraction and simulation to evaluate circuit performance 2.7 Debugging layout and post-layout issues before final design submission
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Specifications and work instructions 4.3 Learning materials 4.4 Tools and equipment

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate:
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	1.1 designed sequential circuits. 1.2 designed arithmetic circuits. 1.3 completed layout and post-layout verification.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: ANALYZE SRAM MEMORY ARCHITECTURE AND 6T BIT-CELL OPERATION	Nominal Duration: 24 hrs.	Unit Code: SICIP-SCD-ICDE-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to analyze SRAM memory architecture and 6T Bit-Cell operation. It specifically includes the tasks of interpreting memory architecture, analyzing 6T SRAM bit-cell operation and interpret SRAM peripherals and stability.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret memory architecture.	1.1 <u>Memory types and architectures</u> are identified according to application requirements. 1.2 SRAM memory organization, arrays, rows, columns, and addressing schemes are interpreted. 1.3 Memory blocks and their functions are identified from design documentation. 1.4 <u>Memory architecture diagrams and specifications</u> are interpreted correctly. 1.5 Memory organization is verified against design requirements.
2. Analyze 6T SRAM bit-cell operation.	2.1 <u>6T SRAM bit-cell structure and components</u> are identified. 2.2 Read, write, and hold operations of the 6T bit-cell are analyzed. 2.3 Word line, bit line, and storage node operations are interpreted. 2.4 <u>Bit-cell simulation results</u> are analyzed to verify functionality. 2.5 Bit-cell operation is evaluated against design specifications.
3. Interpret SRAM peripherals and stability.	3.1 <u>SRAM peripheral circuits</u> and write drivers, are identified. 3.2 <u>Functions of SRAM peripheral circuits</u> are interpreted according to memory operation. 3.3 Read stability, write stability, and hold stability concepts are analyzed. 3.4 SRAM stability and performance reports are interpreted. 3.5 Peripheral operation and stability are verified according to design requirements.

Range of Variables

Variable	Range May include but not limited to:
1. Memory types and architectures	1.1 SRAM 1.2 DRAM 1.3 ROM 1.4 EEPROM 1.5 Flash Memory 1.6 Embedded Memory
2. Memory architecture diagrams and specifications	2.1 Block Diagram 2.2 Memory Map 2.3 Timing Diagram 2.4 Functional Specification 2.5 Address Map 2.6 Design Documentation
3. 6T SRAM bit-cell structure and components	3.1 Six-Transistor (6T) Bit Cell 3.2 PMOS Transistors 3.3 NMOS Transistors 3.4 Cross-Coupled Inverters 3.5 Access Transistors 3.6 Storage Nodes
4. Bit-cell simulation results	4.1 Transient Simulation 4.2 Read Waveform 4.3 Write Waveform 4.4 Hold State Analysis 4.5 Voltage Waveforms 4.6 Simulation Report
5. SRAM peripheral circuits	5.1 Row Decoder 5.2 Column Decoder 5.3 Sense Amplifier 5.4 Write Driver 5.5 Precharge Circuit 5.6 Address Decoder
6. Functions of SRAM peripheral circuits	6.1 Address Decoding 6.2 Bit Line Precharge 6.3 Data Read Operation 6.4 Data Write Operation 6.5 Signal Amplification 6.6 Memory Access Control

Curricular Content Guide

1. Underpinning knowledge	1.1 Fundamentals of semiconductor memory types and memory architecture. 1.2 SRAM organization, memory arrays, addressing
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	schemes, rows, and columns. 1.3 Structure and operation of the 6T SRAM bit-cell. 1.4 Read, write, and hold operations of SRAM memory cells. 1.5 Functions of SRAM peripheral circuits, including sense amplifiers and write drivers. 1.6 SRAM stability concepts, including read, write, and hold stability. 1.7 Memory simulation, performance analysis, and interpretation of design reports. 1.8 Memory design specifications and verification requirements.
2. Underpinning skills	2.1 Interpreting memory architectures, organization, and addressing schemes 2.2 Analyzing 6T SRAM bit-cell structure and operating principles 2.3 Understanding read, write, and hold operations of SRAM bit-cells 2.4 Interpreting word line, bit line, and storage node operations 2.5 Analyzing bit-cell simulation results to verify functional behavior 2.6 Evaluating SRAM stability, including read, write, and hold stability 2.7 Interpreting SRAM performance reports and verifying memory operation against design requirements
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted memory architecture. 1.2 analyzed 6T SRAM bit-cell operation. 1.3 interpreted SRAM peripherals and stability.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DESIGN, LAYOUT AND VERIFY ANALOG CIRCUITS	Nominal Duration: 52 hrs.	Unit Code: SICIP-SCD-ICDE-04-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to design, layout and verify analog circuits. It specifically includes the tasks of applying analog design fundamentals, designing current mirror and differential pair, designing operational amplifiers, applying analog layout techniques, creating matched analog layouts and performing post-layout verification.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply analog design fundamentals.	1.1 <u>Analog circuit requirements</u> are interpreted from the design specification. 1.2 Basic analog components and MOSFET operating regions are applied in circuit design. 1.3 <u>DC and AC operating principles</u> are analyzed according to design requirements. 1.4 <u>Circuit parameters</u> are selected to achieve the required performance. 1.5 Analog design is verified before circuit implementation.
2. Design current mirror and differential pair.	2.1 <u>Current mirror and differential pair circuits</u> are designed using the appropriate EDA tool. 2.2 Device dimensions and bias conditions are selected according to design requirements. 2.3 <u>Circuit simulations</u> are performed to verify functional behavior. 2.4 Simulation results are analyzed and design parameters are optimized. 2.5 Verified circuit schematics are prepared for layout implementation.
3. Design operational amplifiers.	3.1 <u>Operational amplifier specifications</u> are interpreted according to design requirements. 3.2 Operational amplifier schematics are developed using the appropriate EDA tool. 3.3 Gain, bandwidth, stability, and power consumption are verified through simulation. 3.4 <u>Design parameters</u> are optimized based on simulation results. 3.5 Verified operational amplifier design is prepared for layout implementation.
4. Apply analog layout techniques.	4.1 <u>Analog layout planning</u> is carried out according to design rules and specifications.

	4.2 <u>Device placement and routing techniques</u> are applied to minimize parasitic effects. 4.3 Guard rings, shielding, and proper power routing are incorporated into the layout. 4.4 Layout quality is reviewed against analog design requirements. 4.5 Layout is prepared for physical verification.
5. Create matched analog layouts.	5.1 <u>Matched device layouts</u> are created using common-centroid and interdigitated techniques where required. 5.2 Symmetry and matching requirements are maintained throughout the layout. 5.3 <u>Critical analog devices</u> are placed to minimize mismatch and process variation effects. 5.4 Layout is checked for matching accuracy and design rule compliance. 5.5 Completed matched layouts are prepared for post-layout verification.
6. Perform post-layout verification.	6.1 Design Rule Check (DRC) and Layout Versus Schematic (LVS) verification are performed. 6.2 <u>Parasitic extraction and post-layout simulation</u> are conducted using the appropriate EDA tool. 6.3 Simulation results are analyzed to verify circuit performance. 6.4 <u>Layout and performance issues</u> are identified and corrected. 6.5 Verified analog layouts are approved for final design submission.

Range of Variables

Variable	Range
	May include but not limited to:
1. Analog circuit requirements	1.1 Design Specification 1.2 Functional Requirements 1.3 Electrical Specifications 1.4 Performance Requirements 1.5 Design Constraints 1.6 Application Requirements
2. DC and AC operating principles	2.1 DC Biasing 2.2 Operating Point (Q-Point) 2.3 Small-Signal Analysis 2.4 Frequency Response 2.5 Voltage Gain 2.6 Impedance Analysis
3. Circuit parameters	3.1 Supply Voltage (VDD) 3.2 Bias Current 3.3 Transistor W/L Ratio

	3.4 Load Resistance 3.5 Capacitance 3.6 Operating Frequency
4. Current mirror and differential pair circuits	4.1 Current Mirror 4.2 Cascode Current Mirror 4.3 Differential Pair 4.4 Active Load 4.5 Bias Circuit 4.6 Tail Current Source
5. Circuit simulations	5.1 DC Analysis 5.2 AC Analysis 5.3 Transient Analysis 5.4 Operating Point Analysis 5.5 Corner Analysis 5.6 Monte Carlo Simulation
6. Operational amplifier specifications	6.1 Gain 6.2 Bandwidth 6.3 Slew Rate 6.4 Input Offset Voltage 6.5 Power Consumption 6.6 Supply Voltage
7. Design parameters	7.1 Transistor W/L Ratio 7.2 Bias Current 7.3 Compensation Capacitor 7.4 Load Capacitance 7.5 Supply Voltage 7.6 Device Sizing
8. Analog layout planning	8.1 Process Design Kit (PDK) 8.2 Design Rule Manual (DRM) 8.3 Floorplan 8.4 Layout Constraints 8.5 Device Matching Requirements 8.6 Analog Layout Guidelines
9. Device placement and routing techniques	9.1 Common Centroid Placement 9.2 Interdigitated Placement 9.3 Symmetrical Placement 9.4 Metal Routing 9.5 Dummy Devices 9.6 Parasitic Reduction Techniques
10. Matched device layouts	10.1 Common-Centroid Layout 10.2 Interdigitated Layout 10.3 Matched MOSFET Pairs 10.4 Current Mirror Layout 10.5 Differential Pair Layout 10.6 Capacitor Matching Layout
11. Critical analog devices	11.1 Differential Pair 11.2 Current Mirror

	11.3 Input Transistors 11.4 Load Transistors 11.5 Bias Transistors 11.6 Precision Capacitors
12. Parasitic extraction and post-layout simulation	12.1 Parasitic Extraction (PEX) 12.2 RC Extraction 12.3 Post-Layout Simulation 12.4 DC Analysis 12.5 AC Analysis 12.6 Transient Analysis
13. Layout and performance issues	13.1 DRC Violations 13.2 LVS Mismatches 13.3 Parasitic Effects 13.4 Device Mismatch 13.5 Routing Errors 13.6 Performance Degradation

Curricular Content Guide

1. Underpinning knowledge	1.1 Fundamentals of analog circuit design, MOSFET operation, and DC/AC analysis. 1.2 Principles of current mirrors, differential pairs, and operational amplifier design. 1.3 Circuit simulation, performance analysis, and parameter optimization using EDA tools. 1.4 Analog layout techniques, including device placement, routing, shielding, and guard rings. 1.5 Device matching methods such as common-centroid and interdigitated layouts. 1.6 Effects of parasitics, mismatch, and process variation on analog circuit performance. 1.7 Physical verification using DRC, LVS, parasitic extraction, and post-layout simulation. 1.8 Verification of analog layout quality and preparation for final design submission.
2. Underpinning skills	2.1 Applying analog design fundamentals using MOSFET operation, DC, and AC circuit principles 2.2 Performing circuit simulation and optimizing analog circuit performance 2.3 Applying analog layout techniques to minimize parasitic effects and improve reliability 2.4 Creating matched analog layouts using common-centroid and interdigitated techniques 2.5 Maintaining symmetry, device matching, and proper power routing in analog layouts 2.6 Performing DRC, LVS, parasitic extraction, and post-layout simulation for design verification

	2.7 Analyzing and correcting layout and performance issues to achieve final analog design compliance
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 applied analog design fundamentals. 1.2 designed current mirror and differential pair. 1.3 designed operational amplifiers. 1.4 applied analog layout techniques. 1.5 created matched analog layouts. 1.6 performed post-layout verification.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: ADDRESS PARASITIC EFFECTS, RELIABILITY AND ADVANCED PROCESS CONSIDERATIONS	Nominal Duration: 16 hrs.	Unit Code: SICIP-SCD-ICDE-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to address parasitic effects, reliability and advanced process considerations. It specifically includes the tasks of identifying parasitic and reliability effects, applying prevention and mitigation techniques and recognizing advanced process structures.		

Elements and Performance Criteria

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify parasitic and reliability effects.	1.1 Parasitic resistance, capacitance, and coupling effects are identified in IC layouts. 1.2 <u>Reliability issues</u> are recognized. 1.3 Process variations and their impact on circuit performance are analyzed. 1.4 Parasitic extraction and reliability reports are interpreted. 1.5 Design issues related to parasitic effects and reliability are documented.
2. Apply prevention and mitigation techniques.	2.1 <u>Layout techniques</u> are applied to reduce parasitic effects. 2.2 Reliability improvement methods are implemented according to design requirements. 2.3 Shielding, guard rings, and proper routing practices are applied where required. 2.4 <u>Design modifications</u> are performed to improve reliability and performance. 2.5 Updated layouts are verified to ensure compliance with design requirements.
3. Recognize advanced process structures.	3.1 <u>Advanced semiconductor process technologies</u> are identified according to application requirements. 3.2 <u>Process structures</u> are recognized. 3.3 Advanced device structures and layout considerations are interpreted. 3.4 Process technology documents and design rules are reviewed. 3.5 Process selection and technology considerations are related to IC design requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1. Reliability issues	1.1 Electromigration (EM) 1.2 IR Drop 1.3 Latch-Up 1.4 Electrostatic Discharge (ESD) 1.5 Hot Carrier Injection (HCI) 1.6 Bias Temperature Instability (BTI)
2. Layout techniques	2.1 Common-Centroid Layout 2.2 Interdigitated Layout 2.3 Symmetrical Placement 2.4 Short Interconnect Routing 2.5 Dummy Devices 2.6 Parasitic-Aware Routing
3. Design modifications	3.1 Device Resizing 3.2 Routing Optimization 3.3 Layout Compaction 3.4 Power Network Enhancement 3.5 Placement Adjustment 3.6 Constraint Refinement
4. Advanced semiconductor process technologies	4.1 FinFET Technology 4.2 FD-SOI Technology 4.3 CMOS Technology 4.4 Gate-All-Around (GAA) Technology 4.5 Advanced Process Nodes (e.g., 3 nm, 5 nm, 7 nm) 4.6 Silicon-on-Insulator (SOI)
5. Process structures	5.1 FinFET 5.2 FD-SOI 5.3 Multi-Patterning 5.4 Multi-Gate Transistors 5.5 High-k Metal Gate (HKMG) 5.6 EUV Lithography

Curricular Content Guide

1. Underpinning knowledge	1.1 Fundamentals of parasitic resistance, capacitance, and coupling effects in IC layouts. 1.2 Reliability concepts, including electromigration (EM), IR drop, and process variation. 1.3 Methods for minimizing parasitic effects through proper layout techniques. 1.4 Use of shielding, guard rings, and optimized routing for reliable circuit design. 1.5 Interpretation of parasitic extraction and reliability analysis reports.
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	1.6 Advanced semiconductor process technologies and device structures. 1.7 Technology design rules and process considerations for IC implementation. 1.8 Impact of process technology, reliability, and parasitics on IC performance and quality.
2. Underpinning skills	2.1 Analyzing process variations and their impact on circuit performance 2.2 Interpreting parasitic extraction and reliability analysis reports 2.3 Applying layout techniques to reduce parasitic effects and improve performance 2.4 Using shielding, guard rings, and proper routing practices for reliable IC design 2.5 Implementing design modifications to enhance reliability and manufacturability 2.6 Recognizing advanced semiconductor process technologies and device structures 2.7 Interpreting process technology documents and selecting appropriate technologies for IC design
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 identified parasitic and reliability effects. 1.2 applied prevention and mitigation techniques. 1.3 recognized advanced process structures.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration

	2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ANALOG IP LAYOUT PROJECT AND FINAL ASSESSMENT	Nominal Duration: 28 hrs.	Unit Code: SICIP-SCD-ICDE-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform analog IP layout project and final assessment. It specifically includes the tasks of planning and executes the project, preparing documentation and portfolio and presenting and defending design work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan and execute the project.	1.1 Project requirements, specifications, and design objectives are reviewed before implementation. 1.2 Project activities, milestones, and deliverables are planned according to the project schedule. 1.3 <u>Analog circuit design, layout, and verification</u> are executed using the appropriate EDA tools. 1.4 Design Rule Check (DRC), Layout Versus Schematic (LVS), and post-layout verification are completed. 1.5 Final analog IP layout is verified and approved for project completion.
2. Prepare documentation and portfolio.	2.1 Project reports, design files, layouts, and verification results are compiled. 2.2 Design methodology, implementation steps, and verification outcomes are documented clearly. 2.3 <u>Project portfolio</u> is prepared to demonstrate design competencies and completed work. 2.4 Technical documents are reviewed for accuracy, completeness, and consistency. 2.5 Final documentation and portfolio are submitted according to assessment requirements.
3. Present and defend design work.	3.1 Project objectives, design methodology, and implementation results are presented clearly. 3.2 <u>Design decisions and verification results</u> are explained using technical evidence. 3.3 Technical questions and assessment feedback are addressed professionally. 3.4 Design improvements and lessons learned are discussed during the presentation. 3.5 Final project presentation and design defense are completed according to assessment requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1. Analog circuit design, layout, and verification	1.1 Analog Schematic Design 1.2 Analog Layout Design 1.3 Circuit Simulation 1.4 Design Rule Check (DRC) 1.5 Layout Versus Schematic (LVS) 1.6 Post-Layout Simulation
2. Project portfolio	2.1 Design Portfolio 2.2 Circuit Schematics 2.3 Layout Designs 2.4 Verification Results 2.5 Project Documentation 2.6 Presentation Materials
3. Design decisions and verification results	3.1 Design Trade-offs 3.2 Circuit Simulation Results 3.3 DRC/LVS Results 3.4 Post-Layout Verification Results 3.5 Performance Analysis 3.6 Technical Evidence

Curricular Content Guide

1. Underpinning knowledge	1.1 Analog IC design project planning, objectives, milestones, and deliverables. 1.2 Analog circuit design, layout, and verification flow using EDA tools. 1.3 Principles of DRC, LVS, and post-layout verification. 1.4 Technical documentation, design reports, and project portfolio preparation. 1.5 Design verification results and quality assessment methods. 1.6 Professional presentation and technical communication skills. 1.7 Design review, feedback incorporation, and continuous improvement practices. 1.8 Final project submission, design validation, and assessment requirements.
2. Underpinning skills	2.1 Planning analog IC design projects based on specifications, objectives, and project schedules 2.2 Executing analog circuit design, layout, and verification using EDA tools 2.3 Performing DRC, LVS, and post-layout verification for analog IP validation 2.4 Preparing technical reports, design files, layouts, and

	verification documentation 2.5 Documenting design methodology, implementation processes, and verification outcomes 2.6 Developing a professional design portfolio to demonstrate analog IC design competencies 2.7 Defending design work professionally by addressing technical questions, feedback, and design improvements
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 planned and executed the project. 1.2 prepared documentation and portfolio. 1.3 presented and defended design work.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	IC Design and Layout Engineering
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab : 400 sft
- Workshop : 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer/Laptops	01
2.	Printer	01
3.	Multimedia projector	01
4.	High-performance Desktop/Laptop	25
5.	IC Design & Layout EDA Software (Schematic, Layout, Simulation, DRC/LVS)	25
6.	Linux Operating System	25
7.	Central EDA License Server	01
8.	Central File/Storage Server	01
9.	Gigabit Network Switch (48-Port)	01
10.	Enterprise Router	01
11.	Standard Cell & PDK (Process Design Kit) Library	01
12.	SPICE Simulation Environment	25
13.	Schematic Capture Tool	25
14.	IC Layout Design Tool	25
15.	DRC/LVS Verification Tool	25
16.	Layout Parasitic Extraction (LPE) Tool	25
17.	Analog/Mixed-Signal Simulation Tool	25
18.	Waveform Viewer and Debug Tool	25

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on IC Design and Layout Engineering, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.